Linear Algebra (MT-1004)

Homework#15

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Question 1

1. Verify if the vectors $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$ are orthogonal.
2. Given $\mathbf{a} = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$, determine if they are orthogonal.
3. Prove that if $\mathbf{u} \cdot \mathbf{v} = 0$, then $\mathbf{u}$ and $\mathbf{v}$ are orthogonal.
4. **Python Task:** Write a function that checks whether two vectors are orthogonal.

```
def are_orthogonal(u, v, tol=1e-10):
    # Your code here
```